

youkined from <https://github.com/Nicicalu/OST-CN2-Spik>

OSPF (IGP) ECMP load balancing, fast convergence, widely used. IP Port 89 on L3 (no TCP, has own header), Uses 224.0.0.5 (all routers), 224.0.0.6 (all DR,BDR). DR Election: Highest IP prio → if tie, highest Router ID. Priority 0 = ineligible for DR/BDR. No preemption. BDR = same rules, second best candidate Virtual links: Cannot go through more than one stub area, can only run through standard non-backbone areas. (eg. not over stubby areas)	IS-IS (IGP) Widely used by ISPs, Scalable, Fast failure detection & convergence, ECMP. Extendable, Less CPU intensive. Default metric of 10 on most routers → End host devices are called End Systems (ES) IS: Routers are called Intermediate Systems (IS)
ROUTER OPERATION – Establish neighbor adjacencies and exchange LSAs – Build the Link-State Database (LSDB) – Run Dijkstra's SPF algorithm: Intra-area change: SPF recalculation Inter-area change: no SPF needed – ABR handles updates – Build the routing table from SPF results	CONNECTIONLESS NETWORK SERVICES (CLNS) CLNS: ISO Layer 3 datagram service; supports CLNP, ES-IS, IS-IS. CLNP: Connectionless Network Protocol, similar to IP, used in ISO stack ES-IS: host → router discovery protocol IS-IS: router → router. Link-state routing protocol Integrated IS-IS: Allows IP routing with IS-IS
ROUTER TYPES O: Intra-area: Local area → Local area. Type 1 and 2 LSAs B: Intra-area: Local area → Neighboring area. Via LSA 3 through backbone O E1 or O E2: External: Local area → External (eg. BGP), from ASBR O E1 / O N1: Cost = internal + external metrics (def=20) O E2 / O N2: Cost = external metrics. Default external route for OSPF Cost calculation: Reference Bandwidth/(def=100 Mbps)/Interface Bandwidth	NETWORK SERVICE ACCESS POINT (NSAP) Variable in length (up to 20 bytes), identifies node, net, and area from IP: 192.168.1.24 → 49. 0801. 1921. 6808. 1024. 08

AS split into **areas** with 32-bit Area ID (eg. 0, 0.0, 0.0, 0.0 = Area 0)
Backbone Area (Area 0): The OSPF domain, must connect to all other areas (directly/virtual links), must be contiguous (no disjointed segments), should not contain end-user networks, summarizes topology of other areas
Non-Backbone Areas: Connect end users and local resources, All inter-area traffic must transit the backbone

ROUTER TYPES Area Border Router (ABR): Connects two or more OSPF areas, must have 1 interface in backbone, 1 OSPF DB per Area Internal Router (IR): All interfaces belong to the same area (non-backbone) Backbone Router (BR): At least one interface in Area 0, includes ABRs and routers internal to backbone AS Boundary Router (ASBR): Connects to external AS/Network (eg., BGP), Advertises external routes into OSPF, can be in backbone or non-backbone	PACKET TYPES (ALL IN L1 OR L2) All PDUs contain Header with shared common fields + specific routing-related information (Type, Length and Value (TLV)) used to extend protocol IIH (Hello): Build/maintain adjacencies. Includes system ID, holding time, Functions: discover, build, maintain Interval: 10s (default); DIS sends every 3.3s on LANs Multiplier: Missed Hello limit → Holdtime = Interval × Multiplier(def=3) Multicast: L1 01-82-00-02-08-90-14) L2 01-82-00-02-08-90-15) LSP (Link State PDU): Contains topology info including prefixes with costs; unicast on p2p or flooded throughout the area (OSPF: LSA Type 1) CSNP (Complete NSNP): Sent by DIS; lists all known LSAs (OSPF: DBD) PSNP (Partial NSNP): Used to request missing or acknowledge received LSPs
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DESIGN RULES Rule 1: Backbone (Area 0) must be contiguous → no partitions allowed Rule 2: Every non-backbone area must connect to Area 0 LINK STATE ADVERTISEMENT (LSA) TYPES (1) Router-LSA: All routers list directly connected links (so all outgoing interfaces with state and cost) (intra-area) (2) Network-LSA: DR lists all routers and DR in broadcast/multi-access network (intra-area) (3) Summary-LSA: ABR advertises networks from other areas, flooded in all areas (can be "not totally stubby" (inter-area) (4) ASBR-Summary: ABR advertises path to ASBRs. ASBR sends Type 1 + external bit, ABR converts to Type 4 and floods (inter-area) (5) AS-External-LSA: ASBR/ABR advertises external routes (eg., BGP); flooded to all non-stub areas → only normal areas (inter-area) (7) NSAA External: Like Type 5 but inside NSAA . Converted to Type 7 by ABR. Shown in routing table as [O N1 or O N2]	ROUTER TYPES L1 Router: Within one area; all have same LSPDB. No inter-area routing L2 Router: Backbone router, all have same LSPDBs. Routes between areas L1/L2 Router: Acts as both; separates LSPDBs, redistributes between levels ADJACENCIES Backbone must be contiguous: (L2 connection) L1 ↔ L1-L2 ↔ L2. NOTE: L1/L2 routers form both L1 and L2 adjacencies if they are both in the same area. Interface MTU and area key must be identical to become neighbors L1: Area addresses match unless configured otherwise L2: Allside L1, unless router is L1 only, areas must not match POINT-TO-POINT (no DIS) Initialized by receipt of ISHs, exchange p2p IIHs padded to MTU → NSNP. Sent once at adjacency startup LSP: Advertises topology changes (link-state info) PSNP: Acknowledges received LSPs or requests missing ones BROADCAST/MULTI-ACCESS (DIS REQUIRED) Not triggered by ISHs, instead broadcast IIHs with all neighbors MAC's Designated Intermediary System (DIS): Create p2p. Update pseudonode LSPs. Flood LSPs. Similar to DR in OSPF. No backup DIS. DIS election: Highest priority (27) (Cisco def=1) → Highest NSAA (MAC) Preemption: Enabled → higher prio router automatically takes over DIS Role Pseudonode: Carried out by DIS. Multicaccess links. Separate for L1/L2. Fake node IS-IS creates to represent a broadcast LAN Passive interfaces: Advertise network prefixes without adjacency forming
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AREA TYPES (ALLOWED LSA TYPES) Stubby = Blocks external LSA [No 5], Totally = Blocks inter-area LSA [No 3/4] Default rules (eg. 0.0.0.0) from ABR/DR ABR , Cannot connect ASBR [No ASBR] Standard (1,2,3,4,5): Normal Stub Area (1,2,3): [No 4/5] [DR ABR] [No ASBR] Totally Stubby Area (1,2): [No 3/4/5] [DR ABR] [No ASBR] Not-So-Stubby Area (1,2,3,7): [No 4/5], no DR, allows 1 ASBR [LSA 7 → 5] Totally NSAA (1,2,7): [No 3/4/5] [DR ABR] [LSA 7 → 5]	PATH SELECTION Path from L1 always goes through nearest L1/L2 (if no route leaking) Default interface metric (m) > 10 (Lower Metric better) 1) Most specific match (Type: LSP < EGP → internal (1) on origin) 2) L1 intra-area > L2 intra-area > Leaked L2 → L1 (internal m) > L1 external (external m) > L2 external (external m) > Leaked L2 → L1 (external m) 3) Lowest cost 4) Lowest system ID 5) Lowest interface ID ROUTING Summarization occurs when routers enter an IS-IS level (L1 → L2, L2 → L1). LEVEL 1 Intra-area routing only (L1 Area like OSPF Totally Stubby Area) L1: routers install a default route to nearest L1/L2 for inter-area traffic L1-L2: Do not advertise L2 routes into L1 area (unless route leaking active) Set Attached bit to signal L2 connectivity to backbone Distribution Bit: Set to 1 on L2 → L1; blocks re-advertisement L1 → L2. Route-Leaking: injects a more specific route into L1 to improve routing
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SUB-PROTOCOLS Hello Protocol: Neighbor discovery, Parameter negotiation, Maintain logical adjacencies on P2P, P2MP, virtual links. Elect DR/BDR on broadcast and NBMA networks. Continuously send hello packets to maintain bidirectional connectivity, failure to receive = neighbor down (RouterDeadInterval) Database Sync Protocol: Sync LSDB using DBD packets. NEIGHBOR STATES Down: Initial state, no Hello packets exchanged or after RouterDeadInterval 1) Received Hello packet from neighbor but without its own Router ID yet 2) 2-Way: Seen its own Router ID. DR/BDR election is taking place in NBMA ExStart: (DBD) Bi-dir comm; highest Router-ID=master. Determine init Seq Nr Exchange: Exchange of DBD packets (LSA headers) Loading: Missing LSAs are requested. (LSRLSU) Hello: Databases fully synchronized. Normal operating state	ROUTING Each router runs Dijkstra per area; link cost = metric from LSAs (1–65535) – Routes added to RIB/FIB based on computed next hops – ECMP: Modified Dijkstra supports Equal-Cost MultiPath if multiple paths have same cost → routes added with multiple next-hops for load balancing
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PATH SELECTION Path always has to go through backbone 1) Most specific match (CIDR) 2) Static < O > O > I > A > E1 > E2 > N2 3) Lowest link cost (reference bandwidth/interface bandwidth) POSSIBLE FAILURES – No OSPF enabled / Interface not in OSPF / Duplicate router ID – Passive interface that has to be removed / Virtual Link badly configured – Area ID or Area Type mismatch on neighbors – Hello interval or Router dead interval mismatch	POSSIBLE FAILURES – Wrong addressing (NET, Sys-ID, Area ID) / Interface type (P2P, broadcast) – Wrongly configured L1/L1-L2/L2 routers (eg. no L1-L2 as transit) – Auth or MTU mismatch (TLV auth → configure identical IS-IS Auth/MTU) BGP (EGP) Path-vector routing protocol (doesn't contain full topology). Stable, flexible, scalable. Based on TCP. Loop prevention using AS_Path 16BP Routers in same AS, AD=200, more trusted (lower security overhead), AS-Path and Next-hop not modified. iBGP is needed for eBGP because otherwise external IP address might not be known inside the AS with other IGP protocols
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SCALABILITY – iBGP does not re-advertise routes between iBGP peers → full mesh required (cause no loop prevention) – Session count = host × peer × 2 (eg., 5 routers = 10 sessions, 10 = 45) ROUTE REFLECTORS (RR) Solves iBGP full-mesh (split horizon) scaling by selective route refl. Clients only peer with RR; unaware they're clients. Only RR needs special config Clients: iBGP neighbors that rely on the RR for route sharing Non-clients: regular iBGP peers; still have full-mesh among themselves ORIGINATOR ID: stops a router from accepting its own reflected route From non-client: RR advertises to clients only From client: RR advertises to all (clients + non-clients) From eBGP peer: RR advertises to all (clients + non-clients) EBGP Routers in different ASes, AD=20, stricter policy enforcement. Prepends ASN to AS-Path, modifies next-hop IP. Fails if own ASN in AS-Path. TTL=1 AS-Override: Allows reuse of same ASN across different customer sites (eg., Swisscom), rewrites ASN to avoid loop detection Next-hop-self: Router can't install a route if the next hop address is unreachable → next-hop-self changes next hop address to own local addr by BR AUTONOMOUS SYSTEM NUMBERS (ASN) – Unique ID for each AS; required for Internet routing on BGP – Private: [Legacy 16-bit 64512-65535] [32-bit 4200'000'000+294'967'294] PEERING / NEIGHBORS – Two routers with a BGP TCP session (port 179) are called peers/neighbors – Each BGP router is a BGP speaker – Supports CIDR, route aggregation; decisions based on policies/rules PATH ATTRIBUTES Used for route control and policy enforcement Well-known mandatory: Always present (eg., AS-Path, Origin, Next Hop) Well-known discretionary: Optional but recognized by all (eg. Local Pref) Optional transitive: Passed between ASes (eg., Community, Aggregator) Optional non-transitive: Not passed across ASes (eg., MED, Weight) NLRI: Routing table info: prefix, prefix length, and associated path attributes	MESSAGES OPEN: Establishes session; includes version, ASN, Hold Time, BGP Identifier, optional params. Hold Time: Heartbeat in seconds (Cisco def=180s), reset by KEEPALIVE/UPDATE; 0 = session down. BGP Identifier: 32-bit router ID, manually set or highest loopback/address IP; used for loop prevention KEEPALIVE: Every 1/3 of Hold Time; ensures neighbor liveness UPDATE: Advertises new routes, withdraws old ones, or both; includes NLRI (prefix + path attributes); can act as KEEPALIVE NOTIFICATION: Sent on session error (eg. hold timer expired); terminates session immediately NETWORK STATEMENTS Advertise a specific network prefix to BGP peers (only the best is advertised, even if multiple exist). Must exist in the RIB so that it's imported in BGP table Connected: next-hop: 0.0.0.0, origin: attribute 1, weight: 32768 (Cisco) Static/RR: next-hop: NH in RIB, MED: IGP metric, rest same as above
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MESSAGES OPEN: Establishes session; includes version, ASN, Hold Time, BGP Identifier, advertised routes, Hold Time; Heartbeat in seconds (Cisco def=180s), received by KEEPAIVE/UPDATE: 0 = session down, BGP Identifier : 32-bit Router ID, manually set or highest loopback/active IP; used for loop prevention KEEPALIVE : Sent every 1/3 of Hold Time; ensures neighbor liveness UPDATE : Advertises new routes, withdraws old ones, or both; includes NLRI, attributes, and next-hop NOTIFICATION : Sent on session error (e.g. hold timer expired); terminates session immediately	NETWORK STATEMENTS Advertise a specific network prefix to BGP peers (only the best is advertised; even if multiple exist). Must exist in the RIB so that it's imported in BGP table Set origin attribute to origin attribute (weight 32768 (Cisco)) Set BGP next-hop, NH in RIB, MED, IGP metric, next hop as above
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COMMUNITIES 32b optional, transitive tag (eg. ASN: val:UE, 65000:100). Used to mark routes for policy control across ASes. Can be added, modified, or removed at each hop ENTERPRISE CONNECTIVITY OPTIONS Single/Dual: denotes how many links there are Multi-Homed/-Homed: denotes how many ISPs are connected Single-Homed: One ISP, one link (BGP or static). Simple but no redundancy Dual-Homed: Multiple, two links (routers), redundancy within same provider Multihomed: Multiple ISPs, one link. Improved redundancy and routing control, but avoid being a transit – advertise only customer-owned prefixes Dual-Multihomed: Multiple ISPs, two links per ISP. Most redundancy TRAFFIC ENGINEERING (TE) - INBOUND VS OUTBOUND Local Pref: [OUT] [highest] ↑ Lower local pref on backup path Multi-Exit Discriminator (MED): [IN] [lowest] ↓ Higher MED on backup path AS-Path Prepending: [IN] [shortest] ↓ Add multiple own ASN on backup path TE Limitation: can't control IGP, only limited - ISPs may ignore TE with Aggregate: Prefer primary ISP with summarized routes; advertise specific prefixes on backup for failover Transit: One ISP provides reachability to all neighboring destinations Peering: ISPs provide each other reachability to portions of their routing table TRANSIT VS. PEERING Transit: ISP provides full reachability (paid relationship) Peering: ISPs exchange selected routes; equal relationship, usually unpaid AS Path Filtering: to avoid getting transit: ip as-path access-list 10 permit * \$	ROUTING – BGP maintains all received paths per prefix but advertises only the best one – Best path is installed in RIB; recalculated on: – Next-hop reachability change – Interface failure to eBGP peer – Redistribution change – New/withdrawn path received – Influence: – Outbound BGP policy → inbound traffic behavior – Inbound BGP policy → outbound traffic behavior PATH SELECTION (IF PREFIXES ARE THE SAME): 1) Prefer highest Weight (Cisco-specific , local to router) 2) Prefer highest Local Preference (global within AS) 3) Prefer routes originated by the router (only small i in path, NH 0.0.0.0) 4) Prefer shorter AS-path (only length is compared) 5) Prefer lowest origin type (IGP < EGP → internal (1) on origin) 6) Prefer lowest MED (Multi-Ext Discriminator) (also called metric) 7) Prefer external (eBGP) over internal (iBGP) 8) For iBGP: prefer path with lowest IGP metric to next-hop 9) For eBGP: Prefer oldest (more stable) path 10) Prefer lowest BGP router ID 11) Prefer path from lowest neighbor IP address
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STATES			
State	Action	Next (Success)	Next (Failure)
Idle	Waiting for start event	Connect	Idle
Connect	Waiting for TCP handshake	OpenSent	Active
Active	Retrying TCP connection	OpenSent	Connect
OpenSent	Send OPEN, waiting for OPEN	OpenConfirm	Idle
OpenConfirm	Waiting for KEEPALIVE	Established	Idle
Established	Peering active. Exchanging routes.	Established	Idle

COMMUNITIES

32b optional, transitive tag (e.g. ASN:val:ue, 65000:100). Used to mark routes for policy control across ASes. Can be added, modified, or removed at each hop.

ENTERPRISE CONNECTIVITY OPTIONS

Single/Dual: denotes how many links there are

Multi-Homed-Homed: denotes how many ISPs are connected

Multi-Homed-Redundant: One ISP, one link (BGP or static). Simple but no redundancy

Dual-Homed: One ISP, two links (routers). Redundancy within same provider

Multi-Homed: Multiple ISPs, one link. Improved redundancy and routing control, but avoiding being a transit – advertise only customer-owned prefixes

Dual-Multi-Homed: Multiple ISPs, two links per ISP. Most redundancy

Multi-Pref: [OUT] [highest] ↓ Lower local pref on backup path	
Multi-Ext Discriminator (MED): [IN] [lowest] ↑ Higher MED on backup path	
AS Path Prepending: [IN] [shortest] ← Add multiple own ASN on backup path	
AS Path Control: AS controls (control limited) - ISPs may ignore	
AS Path Aggregate: [IN] [summary] ← AS path summarized routes; advertise specific prefixes on backup for fallover	
Transit: One ISP provides reachability to all neighboring destinations	
Peering: ISPs provides reachability to other reachability to portions of their routing tables	
TRANSIT VS. PEERING	
Transit: ISP provides full reachability (paid relationship)	
Peering: ISPs exchange selected routes; equal relationship, usually unpaid	
AS Path Filtering: to avoid getting transit; ip as-path access-list 10 permit *	

INTERNET EXCHANGE POINT (IXP) Facility where networks exchange traffic via BGP peering, routing policies determine which routes are advertised/accepted. Reduces transit costs, latency, and offloads upstream links PRIVATE PEERING Members peer via shared switch fabric and a route server , which distributes routes but stays out of data path (NEXT_HOP unchanged). Simplified setup (one legal contract), minimal policy control; one BGP session to route server PRIVATE PEERING Direct BGP sessions between two parties (2:1), requires one session and legal contract per peer. May use public or private interconnects. Full policy control per neighbor. Examples: Equinix, SwissIX (non-profit) RESOURCE PUBLIC KEY INFRASTRUCTURE (RPKI) – Framework to validate that prefix is legitimately originated by a specific ASN – Prevents route hijacking and accidental mis-originations (route leaks) – RPKI validates origin, not the AS-Path KEY COMPONENTS Trust Anchors (TA): Root CAs of the 5 RIRs; issue certs for resource holders Route Origin Authorization (ROA): Digitally signed object, authorizes ASNs to announce prefix (eg. AS, prefix that AS can originate, max prefix length) RPKI Validators: downloads+verifies+stores Validated ROA Payloads (VRPs) RPKI VALIDATION STATES Valid: Prefix + ASN match ROA Invalid: Prefix found, but ASN mismatch or prefix too long Not Found / Unknown: No matching ROA	VALIDATOR DETAILS – Use TALS (Trust Anchor Locators) to fetch data from RIRs – Validate cryptographic signatures (via X.509 certs with RFC-3779) – Outputs VRPs; invalid objects are discarded – Update frequency: >=24h (recommended), ~30-60min (practice) – RRDP (HTTPS) to fetch RPKI repository data (RFC 8182) replacing rsync – RPKI-RTTR: used to get the validator's already-processed cache of VRPs MONITORING Event tracking, BGP hijack detection, Route leak detection, RPKI status check, Reachability tracking, AS path change tracking, AS path visualization MULTICAST Use-cases 1-to-Many: Streaming, software updates, music-on-hold Use-cases Many-to-Many: Gaming, VR, stock data, group chat Benefits: Efficient bandwidth, low server/CPU load, no redundancy Properties: UDP-based. Apps must handle drops, duplicates, out-of-order Source: Sends to group ID; doesn't need to join Receiver: Must explicitly join group to receive traffic MULTICAST ADDRESS RANGES 224.0.0.0 – 224.0.0.255: Link-local, TTL=1 (not forwarded by routers) (IGMP) 224.0.1.0 – 224.0.1.255: Reserved by IANA, routable (NTP) 224.0.0.0 – 232.0.0.255: Source-Specific Multicast (SSM) 232.0.0.0 – 238.255.255.255: Globally scoped multicast (routable) 239.0.0.0 – 239.255.255.255: Administratively scoped (private space)
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BROADCAST BASICS L3 routes between subnets; L2 floods within same subnet L2 (Bridging): MAC ffff.ffff.ffff switches flood to all ports in VLAN L3 (Routing): 255.255.255.255, local broadcast, not routed. Directed broadcast for specific subnet (eg. 10.1.1.255); can be routed if enabled L2 MULTICAST: MAC HAPPING 1) Set IP: Example multicast IP address: 239.5.5.5 2) Convert to Binary: 239.5.5.5 = 11101111.00000101.00000101.00000101 → Take only the last 23 bits: 0000101.00000101.00000101 3) Map to MAC Prefix: Exed 0100.5E → Map last 23 bits to: 0100.5E05.0505 INTERNET GROUP MANAGEMENT PROTOCOL (IGMP) Purpose: Manages group membership for IPv4 multicast on each segment IGMPv1: Basic join via query-response mechanism, no explicit group leave Router sends membership queries every 60s to 224.0.0.1, removes group after timeout if no report received. Receiver doesn't know multicast source IGMPv2: Adds Leave requests to all interfaces; prunes where no receivers e.g., when someone leaves group ("anyone still interested in group xy?") reducing broadcast overhead. Receiver still doesn't know who the source is IGMPv3: Adds source filtering (Include/Exclude lists). Enables Source-Specific Multicast (SSM) – receiver receives traffic only from selected source(s), no need for Rendezvous Point (RP) anymore. Adds support for application-level access control and filtering. Can also be used in ASM (Any Source Multicast), but mainly with SSM. (224.0.0.2 to all IGMPv3 routers)	IGMP SNOOPING Without snooping: Multicast = broadcast on VLAN With snooping: switch listens to IGMP messages & builds a forwarding table Default: snooping is enabled; switch needs IGMP Querier to operate REVERSE PATH FORWARDING (RPF) To avoid loops, verifies that a multicast packet arrives on the interface that a unicast packet destined for the multicast source would be forwarded out of. RENDEZVOUS POINT (RP) Used in Shared Tree (*,G) setups with PIM-SM, acts as common meeting point. RPF check is performed toward the RP (not the source). Once source is known, routers may switch to a Source Tree (S,G). Triggered by SPT join timer expiration/RPF check confirming better route data exceeding. In IPv6 RPD address can be included in multicast Bb4b[flags]Bb4b[addr]64b[pref]32b
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SHARED TREE VS. SOURCE-BASED TREE Source-Based Tree (S,G): Built and added to mrout table when router receives an (*,G) join/report from IGMP host. s = known source; g = group Shared Tree (*,G): IGMP host sends membership report (IGMP Join). Router adds (*,G) entry to multicast routing table s = anyunknown PROTOCOL INDEPENDENT MULTICAST (PIM) Only between routers. Relies on unicast routing table (RIB) for multicast forwarding decisions and RPF. Protocol-independent. PIM DENSE MODE (PIM-DM) – NO RP PIM-DM relies on IGMP to learn about receivers, prevent unnecessary multicast traffic and trigger pruner behavior. One router elected as IGMP querier, periodically sends IGMP query messages. Essential for IGMP snooping. Push Model: Floods multicast to all interfaces; prunes where no receivers (1) Flooding: Source sends traffic to multicast-enabled links using unicast RIB (2) Distribution Tree: Initially entire network (shared tree rooted at source) (3) Prune Messages: Routers without interested receivers send prunes upstream to remove themselves from the tree (4) State Maintenance: Routers track source, receivers, interfaces to forward/prune per group	ROUTING – BGP maintains all received paths per prefix but advertises only the best one – Best path is installed in RIB; recalculated on: – Next-hop reachability change – Interface failure to eBGP peer – Redistribution change – New/withdrawn path received – Influence: – Outbound BGP policy → inbound traffic behavior – Inbound BGP policy → outbound traffic behavior PATH SELECTION (IF PREFIXES ARE THE SAME): 1) Prefer highest Weight (Cisco-specific , local to router) 2) Prefer highest Local Preference (global within AS) 3) Prefer routes originated by the router (only small i in path, NH 0.0.0.0) 4) Prefer shorter AS-path (only length is compared) 5) Prefer lowest origin type (IGP < EGP → internal (1) on origin) 6) Prefer lowest MED (Multi-Ext Discriminator) (also called metric) 7) Prefer external (eBGP) over internal (iBGP) 8) For iBGP: prefer path with lowest IGP metric to next-hop 9) For eBGP: Prefer oldest (more stable) path 10) Prefer lowest BGP router ID 11) Prefer path from lowest neighbor IP address
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IPV6 PRR Addr can be included in multicast 8b4d[flags]8b4d[addr]64[pre]32[post]	
SHARED TREE VS. SOURCE-BASED TREE	
Source-Based Tree (S,6): Built and added to <i>mroutet</i> table when router receives an (S,6) join/leave from IGMP host. s = known source , 6 = group	
Shared Tree (S,*) : IGMP host sends membership report (IGMP Join) to router adding (*,6) entry to <i>multicast routing table</i> s = any/unknown	
POTENTIAL INDEPENDENT MULTICAST (PIM)	
Only between neighbors. Relies on unicast routing table (RIB) for multicasts forwarding decisions and RPF. Protocol independent.	
PIM DEPEND MODE (PIM-DM) – NO RPF	
PIM-DM bases on IGMP to learn about receivers, prevent unnecessary multicast traffic and trigger prune behavior. One router elected as IGMP querier periodically sends IGMP query messages. Essential for IGMP snooping.	
Push Mode: Floods multicast to all interfaces; enabled links where no receivers	
1) Flooding: Source sends traffic to multicast-enabled links using unicast RIB	
2) Distribution Tree: Initially entire network (shared tree rooted at source)	
Prune: Receivers send prune messages to upstream routers. Upstream routers stream to remove themselves from the tree	
4) State Maintenance: Routers track source, receivers, interfaces to forward	
prune per group	

PIM SPARSE MODE (PIM-SM) – ASM, SSM Push Model: Multicast traffic only sent where requested (explicit join). Works with IGMP to detect interested receivers, uses unicast routing for forwarding (1) Join/Pra: Routers send explicit join/prune messages to request or stop receiving multicast for group (s) to other routers (2) Forwarding: Routers only forward multicast packets for group (s) on interfaces from which explicit joins were received ANY SOURCE MULTICAST (ASM) – Works with IGMPv1 or IGMPv2 → receiver does not know the source – All routers in the multicast domain must know the RP location 1) Receiver sends IGMP Join (*,G) to next hop, which determines the RP for the group and sends a PIM Join (*,G) towards RP, following RPF path 2) Shared tree (s,G) established, informing RP that receiver is interested 3) The source starts sending multicast traffic to its first-hop router (FHR). 4) FHR encaps. multicast packets into PIM Register msg., unicast to RP 5) RP sends a PIM Join (*,G) towards source (FHR), following RPF path towards source. Simultaneously, RP sends Register-Stop to FHR. 6) After receiving Register-Stop, FHR stops encapsulation and forwards multicast traffic directly to the receiver (s,G) 7) The RP forwards multicast traffic down the shared tree towards the LHR. At each hop, routers perform RPF checks 8) LHR receives multicast traffic, performs RPF check towards the source 9) If a more optimal path to the source exists, the LHR sends a PIM Join (s,G) directly towards the source, following the RPF path. 10) After SPT (s,G) is established, traffic flows directly from source to LHR. LHR sends PIM Prune (*,G) towards RP to remove itself from shared tree	ROUTING – BGP maintains all received paths per prefix but advertises only the best one – Best path is installed in RIB; recalculated on: – Next-hop reachability change – Interface failure to eBGP peer – Redistribution change – New/withdrawn path received – Influence: – Outbound BGP policy → inbound traffic behavior – Inbound BGP policy → outbound traffic behavior PATH SELECTION (IF PREFIXES ARE THE SAME): 1) Prefer highest Weight (Cisco-specific , local to router) 2) Prefer highest Local Preference (global within AS) 3) Prefer routes originated by the router (only small i in path, NH 0.0.0.0) 4) Prefer shorter AS-path (only length is compared) 5) Prefer lowest origin type (IGP < EGP → internal (1) on origin) 6) Prefer lowest MED (Multi-Ext Discriminator) (also called metric) 7) Prefer external (eBGP) over internal (iBGP) 8) For iBGP: prefer path with lowest IGP metric to next-hop 9) For eBGP: Prefer oldest (more stable) path 10) Prefer lowest BGP router ID 11) Prefer path from lowest neighbor IP address
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Receiver subscribes using IGMPv3, providing both source (s) and group (g) as to the first-hop router (explicit join) – No RP required → PIM-SSM builds only (s,g) Shortest Path Trees (SPT) – No shared tree (r,g) model used; SSM is source-driven – IMA reserved 224.0.0.0/8 for SSM in IPv4 – Join msgs forwarded hop-by-hop toward source to establish forwarding path – Uses unicast routing table (RPF) to maintain loop-free delivery	
PIM SPARSE-DENSE MODE	
PIM Sparse Mode: Pull model – multicast traffic is forwarded only on request PIM Dense Mode: Push model – traffic is flooded everywhere, then pruned PIM Sparse Mode: Supports both per multicast group, depends on RP availability	
PIM SPARSE MODE VS PIM DENSE MODE	
DM: Small tightly coupled networks where most devices need multicast SM: Large-scale/distributed env where multicast receivers are fewspread out	
NETWORK DESIGN	
Pillars: Scalability, Speed, Availability, Security, Manageability → overall cost	
AVAILABILITY	
MTTD: Mean Time To Detect MTRR: Mean Time To Repair MTBF: Mean Time Between Failures MTBF combined: $(\sum_{i=1}^n 1/MTBF_i)^{-1}$ Availability: $MTBF / (MTBF + MTRR)$	MTTI: Mean Time To Identify MTRRS: MTT Restore Service MTBSI: Mean Service Incidents MTBF parallel: $\sum_{i=1}^n MTBF_i$ Better Av: $MTRR \rightarrow 0 \wedge MTBF \rightarrow \infty$
REDUNDANCY	

WAN
Connects remote LANs via S/Ps for data/voice/video; key needs: bandwidth, control, design, resilience, mgmt. **Requirements:**
Bandwidth: Plan for normal load, peak usage, reserved capacity for VoIP
Control/Security: Reliance on the provider → encryption and segmentation
Availability: Use redundancy, diverse paths, SLAs to reduce outage impact
Management:
In-band: Manage devices over the production WAN
Out-of-band: Use separate management path for recovery/emergencies

PRIVATE WAN
Point-to-Point: Leased L2 line (Ethernet), monthly fee, private circuit, fast
Dark Fiber: Physical fiber lease; costly; ISPs prefer selling lambdas
Coarse Wavelength Division Multiplexing (CWDM): x16, cheaper, 120km, passive MUX, use mirrors to aggregate/separate signals (analog)
Dense Wavelength Division Multiplexing (DWDM): x80, 10GB/s, 1000km, active MUX, more precise, uses software ("smarter"), wideband
connection-oriented: Predelineated path, packet IDs (ATM, Frame Relay)
Connectionless: No setup, full address in each packet (Ethernet, MPLS VPN)
Circuit-Switched: Requires signaling protocol to establish e2e circuit
Packet-Switched: Expects network to know how to forward packets

PUBLIC WAN
Site-to-Site VPN: Fixed locations, devices locked to IP addr
Remote Access VPN: Changing locations, devices not locked to IP addr
Software-Defined WAN (SDWAN): connects LANs across large distances using controlling software that works with a variety of networking hardware.

MULTI PROTOCOL LABEL SWITCHING (MPLS)
– Advantage of eBGP at PE-CE: No mutual redistrib., same routing process
– Control plane (e.g. OSPF) → to learn labels
– iBGP to exchange NLRI (RD, RT/v4 Pref, NextHop&VPN Lb) between PE
– Traffic encrypted flowing over MPLS L3VPN backbone (e.g. Bank)

Label Switched Router (LSR): Forwarded labeled packets
Label Switched Path (LSP): Pre-determined path across MPLS network
Unicast Reverse Path Forwarding (uRPF): checks source of each packet & verifies that source is in routing table
imp-nul0: networks are directly connected, no more label switching
Virtual Private Wire Service (VPWS): Pseudowire (PW); No MAC learning at PE level; P2P L2 VPN, allows two sites to connect as if directly linked
Virtual Private LAN Service (VPLS): Emulates LAN segment across MPLS backbone, provides multipoint L2 connectivity; MAC learning at the PE level

CONFIG
1) RIB entry to 10.3.3.0/24 leads a next-hop IP address of 192.168.12.2
2) PE1 compares that entry to the list of interface IPs on each peer and finds nei with IP 192.168.12.2. The LDP ID (LID) of this peer is 2.2.2.2
3) PE1 notes that for 10.3.3.0, LID contains one local and two remote labels
4) Among the known labels, one was learned from 2.2.2.2, with label 22

PE1# show ip route 1 inc 10.3.3.0
1 10.3.3.0 [90/2835456] via 192.168.12.2, 00:05:08, Serial0/0/1/0
PE1#
PE1# show mpls ldp bindings 10.3.3.0 24
Lib entry: 10.3.3.0/24, rev 28
Local bindings: Lsr: 2.2.2.2, 0, label: 22
remote binding: Lsr: 4.4.4.4, 0, label: 86
PE1#
PE1# show mpls ldp nei 4
Peer LDP Ident: 2.2.2.2; Local LDP Ident 1.1.1.1;0
TCP connection: 2.2.2.2.2.34326 -> 1.1.1.1.6446
State: Oper; Hgss ssent/rcvd: 37/33; Downstream
2 Up time: 0:02:57
LDP discovery sources:
Serial0/0/1, Src IP addr: 192.168.12.2
Addresses bound to LDP Ident: 1
1 192.168.12.2 2.2.2.2 192.168.12.0 192.168.24.0

ROUTER TYPES
Customer Edge (CE): No knowledge of MPLS, no labels; connected to **Provider Edge (PE):** On edge of MPLS net; runs BGP & LDP; uses VRFs
Provider or LSR (P): Inside MPLS VPN, no CE connection; forwards labels

DATABASES, PLANES
Control Plane: Builds routing/labeling (RIB, LID)
Data Plane: Forwards packets (FIB, LFIB); pushes/swaps/pops labels
Class of Service Forwarding (CoS): (Cisco) LFIB without labels → FIB? **Routing Information Base (RIB):** Learned prefixes from routing protocols
Forwarding IB (FIB): Built from RIB; only best routes for forwarding
Label IB (LIB): All label mappings; No BGP prefixes; 1 label per prefix
Label Forwarding IB (LFIB): Built from LIB; used for actual forwarding decisions. (LFIB only contains currently best LSP decision: Routing Protocol)

MPLS HEADER
4-byte header used by Wi. With MPLS VPN, there will be a stack of two headers
Label IB: active MPLS label
EXP (3b): CoS/QoS, Now called Traffic Class (TC)
S-bit (1b): Bottom of label stack indicator, 1 = True = last label before IP header
TTL (8b): time-to-live (equal to IP TTL)

TTL
– Ingress PE decrements IP TTL field & copies IP TTL field into new MPLS TTL
– P routers decrements MPLS TTL
– Egress PE decrements MPLS TTL, pops final MPLS header, copies IP TTL
– If provider doesn't want to expose MPLS network: disable **MPLS TTL propagation** on PE, PE sets MPLS TTL=255, egress PE leaves original IP TTL unchanged → new network appears as single router hop from TTL perspective

LABEL DISTRIBUTION PROTOCOL (LDP) - CONTROL PLANE
– Distributes labels to neighbors, triggered by new IP route in FIB
– **Hello messages** sent via UDP (Port 646) to 224.0.0.2 to discover neighbors
– TCP (646) connection used to exchange label bindings (prefix to local lib)
– Router advertises all local bindings after TCP session is up
– Label mapping used to build LIB → LFIB
– LDP router ID must be reachable (via routing table)
– Each router manages local labels independently
– Relies on underlying routing protocol for best route & loop prevention

MPLS L3 DATA PLANE
Outer label: Transport label (LDP); identifies LSP between ingress/egress PE
Inner label: Transport label (BGP); identifies customer VRF
Push: Ingress PE: classify and label packets
Swap: P router; replaces label, forwards based on new VRF
Pop: Egress PE removes label; sends original packet to CE
Penultimate Hop Popping (PHP): penultimate router removes the outer MPLS label before forwarding to egress PE, default enabled, ISPs disable it

VIRTUAL ROUTING AND FORWARDING (VRF) TABLES
– Virtual router inside a PE. Maintains isolated RIB + FIB and separate routing process per CE (VPN isolation)
– Exists per MPLS-aware PE router; one per attached customer

ROUTE DISTINGUISHER (RD) VS. ROUTE TARGET (RT)
Route Distinguisher (RD): *Uniquely identifies* VPN routes (per VRF) → allows same IP prefix to be used in different VPNs. Can be IPv4 or ASN.
– **Form:** VPNv4 NLRI (MP-BGP): RD-IPv4 prefix (12 bytes: RD 8b, IP 4b)
Route Target (RT): Controls route **import/export** between VRFs. Used as **extended BGP community path attributes**. Typically in the form **ASN:nn** or **IP:nn**, e.g., 65000:100, 1:10
How RTs Work: Route is tagged with an RT when advertised by BGP. Other VRFs import the route if the RT matches their import policy. Allows overlapping or shared connectivity between tenants (e.g., shared services).

OVERLAY TECHNOLOGIES
Overlay network (virtual) sits on top of the **underlay network** (physical).
Generic Routing Encapsulation (GRE): Encapsulates data packets, unencrypted, P2P. Enables usage of unsupported protocols. Carries passenger protocol/original packet inside carrier/transport IP header routed by underlay

SEGMENT ROUTING (SR)
Source routing: Sender defines full path using Segment (= Instruction) List
Segment Identifier (SID): Each SID = 1 instruction (e.g., forward via ECMP, specific face, or to a service)
Segment List: Ordered SID list carried in packet header; defines full route
State in packet: No per-flow state, intermediate nodes follow SID instructions
No new control plane: Uses existing protocols (OSPF, IS-IS, BGP) with extensions; no LDP or RSVP-TE needed
Data plane: Can operate over MPLS or IPv6
Simple but powerful: Enables TE, fast reroute, policy routing

SEGMENT LIST OPERATIONS
Push: Insert SIDs into packet; set active SID (top of list)
Continue: Active SID not yet completed; keep processing it
Next: Current SID completed; activate next SID in list

SEGMENT SIGNIFICANCE
Global Segments: Known and supported by all SR nodes in the domain, installed in forwarding tables across the network (e.g., "Forward packet according to shortest path to Node1"). Defined in the SR Global Block (SRGB) and should be consistent across all nodes
Local Segments: Defined and installed only on originating node, not forwarded by others, but must be understood network-wide (e.g., "Forward packet on interface to Node2"). Defined in the SR Local Block (SRLB) and are specific to the local SR node.

CONTROL PLANE SEGMENT TYPES
IGP Prefix Segment: Global SID tied to IGP prefix (multi-hop); all nodes install forwarding entries: "steer traffic along ECMP-aware shortest path"
IGP Node Segment: Global SID for a specific node (shortest-path forwarding)
IGP Adjacency Segment: Local SID for a specific link; traffic sent to nearest IGP Adjacency Segment
L2 Adjacency SID: Local SID for Layer-2 segment (e.g., Ethernet link)
Combining Segments: End-to-end paths can mix IGP and BGP segments. Traffic to BGP Anycast → more ECMP in data centers

SR-MPLS
– Reuses existing MPLS data plane → no hardware change needed
– Segments = MPLS labels; Segment List = label stack (top = active)
– Segments distributed via IGP/BGP; no LDP necessary (but possible)
– Supports both IPv4 and IPv6 networks
– PUSH= PUSH, CONTINUE= SWAP, NEXT= POP

BENEFITS OF SEGMENT ROUTING
Benefits: Simplification (removes protocols, simple operations, admin and mgmt), enhanced Traffic eng. (Delay, Bandwidth, Packet Loss, TE metric, Controller, Source-Node), Seamless deployment, Robust
Source Routing: Labels distributed inline with centralized optimization
Traffic Engineering (TE): Optimizes network performance by analyzing and controlling data flow to reduce congestion and improve QoS

DRAWBACKS OF TRADITIONAL NETWORKS
Control Plane: LDP/RSVP-TE adds complexity
Scalability: Per-flow/path limits growth; LSP, signaling overhead increase fast

VIRTUAL EXTENSIBLE LOCAL AREA NETWORK (VXLAN)
Data plane technology which encapsulates Ethernet frames in UDP data packets to tunnel layer 2 frames over a layer 3 network. The underlay network is unaware of the overlay.
Issues of L2: STP. Max amount of VLANs (4094). Large MAC Address tables
VXLAN: Tunnels Ethernet (Layer 2) over IP using MAC-in-UDP encapsulation (Port 4789). For flexible and scalable network segmentation.
VXLAN Identifier (VNI): 24-bit identifier (Up to 16 million segments) that defines the VXLAN broadcast domain.
Virtual Tunnel Endpoint (VTEP): Device (switch, router, or host) responsible for encapsulating/de-encapsulating VXLAN traffic.
Network Virtual Interface (NVE): Logical interface on a VTEP used for VXLAN tunnel operations.

FRAME FORMAT
– Original Ethernet frame → VXLAN Header → UDP → Outer IP Header
– The VXLAN header contains the 24-bit VNIID and flags.
– Outer headers allow Layer 2 frames to traverse IP underlay networks.

VIRTUAL NETWORK IDENTIFIER (VNI)
– 24-bit VXLAN Network Identifier uniquely defines VXLAN segments.
– Replaces traditional VLAN IDs (12-bit)
– Used by VTEPs to map traffic into corresponding Layer 2 domains.

VXLAN TUNNEL ENDPOINT (VTEP)
Connects the overlay (VXLAN) and underlay (IP) networks. A VTEP can have multiple VNI interfaces, but they associate with the same VTEP IP interface.
Software VTEP: Located on hypervisors using virtual switches.
Hardware VTEP: Located on routers/switches with ASICs for performance.
VTEP Interface: Connects to the underlay network, handles encapsulation.
VNI Interface: Virtual interface per segment (like SVI); handles segregation of Layer 2 domains.

MAC ADDRESS LEARNING
Each VTEP maintains a VXLAN mapping table linking destination MAC addresses to remote VTEP IPs.
Data plane: reactive, Flood and learn, MAC learned from incoming traffic, BUM via ingress replication or multicast underlay. No control-plane needed
Control plane: proactive, MP-BGP, EVPN/MAC-to-VTEP advertised via BGP. No flooding needed for known MACs, ARP suppression, silent host handling
Flood and learn: Host H1 sends ARP request, switches learn H1's MAC. ARP request is flooded to H2 → H2 responds; switches learn H2's MAC.
Static VXLAN: Manual MAC-to-VTEP mappings. Sends copy of BUM traffic to all participating VTEPs. Doesn't scale well; BUM traffic is inefficient.
Multicast VXLAN: VTEPs join multicast groups per VNI. Scales better, offloads BUM replication. 20+ VTEPs = too much traffic, doesn't scale well
MP-BGP EVPN: Modern solution using BGP as control plane. Dynamically learns MAC/IP info.

ETHERNET VIRTUAL PRIVATE NETWORK (EVPN)
Overcome flood-and-learn limitations, doesn't rely on data plane learning, utilizes router control plane MP-BGP, works with different encapsulation techniques (VXLAN, MPLS), excellent scalability, L2 and L3 Support.

LAYER 2
L2 bridging across L3 networks
BGP Control Plane: Distributes MAC info (no flooding → efficiency)
VXLAN Overlay: Encapsulates L2 in L3 UDP (data plane)
Multi-Tenancy: via VNI segmentation
Redundancy: All-active multihoming, ECMP, fast convergence

USE CASES
– Multi-tenant datacenter interconnections (DCI)
– Extending L2 over WAN between remote sites
– Scalable, segmented L2 fabrics

AUTO-DISCOVERY VIA ROUTE REFLECTORS
– RR reflects EVPN routes to other PEs
– RR doesn't participate in EVPN or pseudowires
– RR needs only **address-family l2vpn evpn**
– L2VPN RIB stores endpoint/VFI info for control plane
– BGP UPDATE from spines contains **ORIGINATOR_ID** (origin leaf)

HOST DETECTION
Host connects to VTEP → MAC learned locally
– VTEP advertises MAC + L2VNI via BGP EVPN
– MAC learning follows normal Ethernet semantics

INGRESS REPLICATION (IR)
Forwarding BUM traffic. When Multicast underlay network is not used, IR handles multi-destination traffic (ARP → unicast). Ingress device replicates BUM packets, sends them as separate unicast to egress devices.

EARLY ARP DISCOVERY (ARP SUPPRESSION)
– Avoids flooding ARP requests
– VTEP queries control plane for MAC/IP/VNI mapping
– If known → direct unicast (no broadcast)
Silent Host Flow (Fallback)
– If IP/MAC unknown → ARP sent via ingress replication
– Replicated ARP request goes to remote VTEPs
– Only correct host responds → update reflected to all VTEPs
– Future traffic uses updated BGP mapping

LAYER 3
By adding L3 features, data routing efficiency + smooth connections improve

EVPN ROUTE TYPES
1) Ethernet Auto-Discovery Route (3) Inclusive Multicast Route (4) Ethernet Ingress Tunnel Route (5) Selective Multicast Ethernet Tag Route (7) IGMP Join Synchrony Route (8) IGMP Leave Synchrony Route
Type 2 – Host Advertisement: Advertises host MAC (mandatory), optionally IP, along with L2VNI and optionally L3VNI. Used for MAC learning, ARP suppression, and host mobility. Sent when host connects to VTEP.

[2]: Route Type [0]: Eth. Segment ID [0]: Eth. Tag ID [48]: MAC Addr len [a2b9.4605.948d]: MAC [32]: IP Addr len [10.1.1.1]: IP

RT: L2VNI (VLAN) RT: L3VNI (VRF) Overlay Encapsulation: 8 - VXLAN L2VNI L3VNI Router MAC of Remote VTEP Remote VTEP IP

LEAF-SW show bwp l2vpn evpn 10.1.1.1
BGP routing table information for VRF default, address family L2VPN EVPN
Route Distinguisher: 10.0.0.3:32777
BGP routing table entry for [2]: [0]: [0]: [48]: [a2b9.4605.948d]: [32]: [10.1.1.1]/772, version 424
Paths: (2 available, best #2)
Flags: (0x00002) on xmit-1st, is not in L2rib/evpn, is not in HW
Advertised path: 1
Path type: Internal, path is valid, is best path, no labeled nexthop
AS-Path: none, path sourced internal to AS
10.0.1.101 (metric 81) from 10.0.0.1 (10.0.0.1)
Origin IGP, MED 0, LocalPref 100, weight 0
Received label 100010 100999
Extcommunity: RT:645080:100010 RT:645080:100999 ENCAP:8
Router MAC:00d0.0a00.0168
Originator: 10.0.0.3 Cluster List: 10.0.0.1

Type 5 – Subnet Advertisement: Advertises IP prefix + prefix length to L3VNI. Used for inter-subnet routing. VTEP redistributes connected/static/dynamic IP routes. Additional attributes: L3VNI, extended communities.

[5]: Route Type [0]: Eth. Segment ID [0]: Eth. Tag ID [24]: IP Pref. len [198.51.100.0]: IP Prefix [6.0.0.0]: GW IP

L3VNI RT: L3VNI (VLAN) Overlay Encapsulation: 8 - VXLAN Router MAC of Remote VTEP Remote VTEP IP

LEAF-SW show bwp l2vpn evpn route-type 5
BGP routing table information for VRF default, address family L2VPN EVPN
Route Distinguisher: 10.0.0.3:3
BGP routing table entry for [5]: [0]: [0]: [24]: [198.51.100.0]: [0.0.0.0]/224, version 6421
Paths: (2 available, best #2)
Flags: (0x00002) on xmit-1st, is not in L2rib/evpn, is not in HW
Advertised path: 1
Path type: Internal, path is valid, not best reason: Neighbor
Address: no labeled nexthop
AS-Path: none, path sourced internal to AS
203.0.113.1 (metric 81) from 192.0.2.12 (192.0.2.2)
Origin incomplete, MED 0, LocalPref 100, weight 0
Received label 10001
Extcommunity: RT:645080:10001 Router MAC:00d0.0a00.0168
Originator: 192.0.2.3 Cluster List: 192.0.2.2

HOST DETECTION
– Host sends ARP/ND to local VTEP
– VTEP learns MAC/L2VNI and IP/L3VNI
– Info is advertised in EVPN (control plane)

HOST DELETION & EVPN
Host Deletion: When a host detaches, its ARP (default: 1500s) and MAC entry (default: 1800s) time out on the VTEP. Upon aging, the VTEP withdraws the host's MAC/L2VNI and IP/L3VNI advertisements.
Host Move: When a host moves to a new VTEP, the new VTEP advertises updated reachability with a higher move sequence number. The old VTEP withdraws its entry, completing the migration.

VIRTUAL ROUTING AND FORWARDING (VRF)
– Supports independent policies per tenant
– Key for scaling and multi-customer separation
DISTRIBUTED ANYCAST GATEWAY (DAG)
– Same gateway IP+MAC on all VTEPs

– Enables local default gateway for hosts in MPLS network with IRB
– Supports mobility + optimal forwarding across BGP EVPN fabric

INTEGRATED ROUTING AND BRIDGING (IRB)
– Enables inter-VLAN routing inside EVPN
– Avoids central gateway → "no traffic tromboning"

ASYMMETRIC IRB (L2)
– Routing only ingress, bridging on egress
– VXLAN uses destination VNI in both directions
– One L2 VNI per VLAN/Subnet
– Simple config, but requires all VLANs/VNIs on all VTEPs (scaling issues)

Diagram 1: Leaf 1 (ingress) routes + bridges VNI 10010 VNI 10020 → Spine → Leaf 2 (egress) bridges only VNI 10020. VM A VNI 10010 → Leaf 1 → Spine → Leaf 2 → VM B VNI 10020. VXLAN encap L2 VNI 10020.

1) VM A → Leaf 1 in VNI 10010 (frame to anycast gateway)
2) Leaf 1 routes locally. VNI 10010 → L3 loopback, MAC rewrite)
3) Leaf 1 encapsulates with destination L2 VNI 10020 → spine → Leaf 2
4) Leaf 2 decapsulates and bridges in VNI 10020 to VM B → no routing

SYMMETRIC IRB (L2 + L3)
– Routing/bridging on ingress + egress VTEPs
– Uses L3 Transit VNI (same in both directions), One L3 VNI per VRF (Tenant)
– Scales well, clear separation of MAC and IP
– Requires L3 connectivity between all destination VTEPs for Type 2 routing (Done by configuring Type 5 routing)

Diagram 2: Leaf 1 (ingress) routes + bridges VNI 10010 VNI 50000 → Spine → Leaf 2 (egress) routes + bridges VNI 50000 VNI 10020. VM A VNI 10010 → Leaf 1 → Spine → Leaf 2 → VM B VNI 10020. VXLAN encap L3 VNI 50000.

1) VM A → Leaf 1 in VNI 10010 (frame to anycast gateway)
2) Leaf 1 routes locally into the L3 VNI (tenant VRF transit)
3) Encapsulation carries the L3 VNI 50000 → Leaf 2
4) Leaf 2 decapsulates, looks up dest IP in the VRF, routes into VNI 10020
5) Leaf 2 bridges to VM B on the local subnet

MULTI-PROTOCOL BGP (MP-BGP)
Address Family Identifier (AFI): Category of information being carried
Subsequent AFI (SAFI): Narrows down the specific type within that category
EVPN NLRI: Uses MP-BGP with specific AFI/SAFI. Supports multiple route types and attributes, unsupported routes are dropped by BGP

– Enables protocol-based VTEP discovery and host reachability via control-plane learning. Reduces flooding by replacing data-plane learning
– Extends BGP with multiprotocol capabilities (AFI/SAFI)
– Uses **MP_REACH_NLRI** and **MP_UNREACH_NLRI** for route advertisement and withdrawal (advertises VPNv4 between PEs in MPLS)
– Uses the **VPNv4 NLRI** to distinguish between duplicate IPv4 prefixes

BGP CONTROL PLANE
– PEs learn MACs from local CE's (data plane)
– MACs advertised via BGP (control plane)
– Uses Route Distinguishers and MPLS labels
– Remote PEs update L2 RIB/FIB with MAC and next-hop info
– Enables seamless L2 across IP/MPLS backbone

CDN
Origin Server: Central content source (original files), usually in a datacenter
Edge / CDN Server: Also called **Point of Presence (POP)**. Geographically distributed, caches content
Embedded POP: Deployed directly inside ISP's local network
DNS Infrastructure: Directs users to optimal edge server (e.g. Geo-Routing)

KEY BENEFITS
Latency Reduction: Nearby edge servers reduce round-trip time
Availability: Failover and redundancy in case of node failure
Scalability: Handles traffic spikes via load balancing
Cost Optimization: Reduces backend and transit load on origin
DDoS Protection: Edge servers absorb attacks → not all traffic on one server
Global Load Reduction: Less long-distance traffic across the Internet

REQUEST ROUTING TECHNIQUES
– Decides which edge server should serve a client request
– Goal: Best performance (e.g. proximity, load, responsiveness)

DNS-BASED GEO-ROUTING
– Each edge has a unique IP
– DNS server picks closest/optimal edge server based on:
– Resolver IP/location (not user) → can cause wrong choice)
– DNS (MaxMind, IP2Location), load, latency, business rules
– **EDNS0** and Client Subnet Extension (ECS):
– Resolver includes part of client IP in DNS request (e.g. /24 subnet)
– Authoritative DNS makes better decision based on actual client region
– Improves accuracy without revealing full IP, increases caching complexity

ANYCAST WITH BGP
– Same IP (e.g. 7.7.7.7) advertised from multiple locations
– BGP routing decides which path is "best" (AS-path, local pref, etc.)
– No DNS logic or per-client decision – pure BGP convergence
– **Pros:** Fast failover, simple, no app logic needed
– **Cons:** Control BGP, BGP + best latency, route flapping risk

MANAGING TRAFFIC
Anycast: Modify BGP attributes/withdraw IP prefix. Not precise
Software: Shifting via Encapsulation tunnels (GRE)

LOAD BALANCING
Load balancing device: Efficient, reliable, High cost, single point of failure
ECMP: ToF router balances traffic. Simple, cheap, Adding/removing server changes hashing pool, re-hashing breaks TCP connections if: GRE tunnel

HTTP CACHING & HEADERS
Caching is controlled via HTTP headers between clients, proxies, and servers
Cache-Control: Main directive (**no-cache, no-store, max-age, must-revalidate, etc.**)
Expires: Absolute expiration time (older method, replaced by **Cache-Control**)
ETag: Validator tag (version/hash), used with **If-None-Match**
Last-Modified: Timestamp used with **If-Modified-Since** for validation
Age: Time (in seconds) since response was fetched from origin
Validation: Client uses **ETag** or **Last-Modified**; server → 304 if unchanged

QoS
Internet is best effort: No guarantees, no QoS; all traffic treated equally (net neutrality); simple, scalable, but no delivery/order assurance or prioritization
Quality of Experience (QoE): Perceived service quality from user perspective
Queue Pining: Keeps flow on a fixed path to prevent oscillation (don't switch immediately to "better" path)
Routing-based: Path selection; QoS-aware routing; Choose best path
Node-based: Packet treatment; Congestion mgmt; queuing/scheduling/policing/shaping; "What happens to packets at each hop?"
PAK-Priority: System queues (routers) that prioritizes routing protocol packets

4-Class Model

4-Class Model	8-Class Model	12-Class Model
Real Time	Voice	Voice
	Interactive Video	Real-Time Interactive
		Multimedia Conferencing
	Streaming Video	Broadcast Video
		Multimedia Streaming
Signaling or Control	Signaling	Signaling
	Network Control	Network Control
		Network Management
Critical Data	Critical Data	Transactional Data
		Bulk Data
Best Effort	Best Effort	Best Effort
	Scavenger	Scavenger

NETWORK PERFORMANCE METRICS
Bandwidth [Gbit/s]: Maximum transfer capacity of a link (width of pipe)
Throughput [Gbit/s]: Successfully delivered data (water that reached dest)
Latency / Delay [ms]: Time for packets to travel src → dest (length of pipe)
End-to-End Delay: Total time sender to receiver
One-Way Delay: From first bit sent to last bit received
Component: Transmission delay (time to push onto link), **Processing delay** (lookup, queuing), **Propagation delay** (physical travel time)
Jitter [ms]: Variation in delay between packets, caused by re-routing/queuing (Voip>30ms), Calc: highest - lowest delay. Doesn't impact TCP
Packet Loss Ratio (PLR) [%]: Dropped packets due to congestion or errors

QUEUEING
Incoming buffer: Usually only transit, cannot be filled up in case of a CPU overload; leads to drops
Outgoing buffer: Filled up with packets waiting for transmission, queue management is focused on the outgoing buffer

ALGORITHMS
First-In-First-Out (FIFO): Basic, no prioritization
Priority Queuing (PQ): Multiple queues, serve highest first; others may starve
Round-Robin (RR): One packet per queue in turn (fair, but ignores priority)
Weighted Fair Queuing (WFQ): RR with weights, n packets per queue
Class-Based WFQ (CBWFQ): WFQ with user-defined classes (logical queues), queues limits, max bandwidth guaranteed or max % of bandwidth
Low Latency Queuing (LLQ): Adds strict priority queue (priority class) to CBWFQ for delay-sensitive traffic (e.g. voice) (based on IP Precedence, DSCP, src, port, protocol...)

QUEUE MANAGEMENT
tail Drop: Drops packets when queue full; huge interruption of traffic → same as no connectivity
TCP Global Sync: Many TCP flows back off and restart simultaneously (AIMD) → link underutilization
TCP Starvation: TCP slows down after drops (congestion control), UDP doesn't → queues filled with UDP TCP squeezed out
Random Early Detection (RED): Random early drops before full queue to prevent global sync and TCP collapse. Dropped TCP segments cause TCP sessions to reduce their window sizes
Weighted RED (WRED): RED + DSCP/EXP-based drop logic, prioritizes higher-marked traffic, max thresh → all packets dropped

POLICING VS. SHAPING
Policing (Inbound mostly): Drops packets that exceed configured rate limits
Shaping (Outbound): Buffers packets to smooth traffic bursts

QoS MODELS
BEST EFFORT
No guarantees, all traffic treated equally (follow Internet neutrality)
INTEGRATED SERVICES (INTSERV)
– End-to-end QoS (guaranteed for specific apps)
– Per-flow resource reservation
– Precise but not scalable (uses RSVP)
– App informs network of traffic traits, requests service before sending data
DIFFERENTIATED SERVICES (DIFFSERV)
– Scalable approach using marking. No hard guarantees, Cost-effective
– Divides traffic into classes based on business requirements
– Routers are configured with policy on how to prioritize classes
Marking (DSCP)
Differentiated Services Code Point (DSCP): 6-bit field in the IP header used to classify and manage network traffic
L3 Marking: ToS byte → DSCP (6 bits) + IP Precedence (3 bits) [overlapping]
DSCP: (6-bit in IP header) marks packets for QoS; classifies traffic
L2 Marking: Dot1q header → 802.1p/CoS bits
EXP: (3-bit in MPLS label) serves same purpose within MPLS networks; often mapped from DSCP

Behavior (PHB)
Per-Hop Behavior (PHB): Defines what routers actually do
Class Selector (CS): Compat with old IP Precedence, simple prio levels
Assured Forwarding (AF): Multiple classes + drop priorities
Expedited Forwarding (EF): Very high prio, e.g. VoIP

MODULAR QoS CLI (MQC)
Class Map: Define traffic classes (e.g., match voice or video)
Policy Map: Define actions for each class (e.g., limit, shape, priority)
Service Policy: Apply policies to interfaces or directions (in/out)

OTHER INFOS
AD: Inter-protocol choice (e.g., OSPF vs RIP) → lower wins.
Cost/Metric: Intra-protocol choice (e.g., OSPF path A vs B) → lower wins
Routing Preference Order (across protocols):
– Most specific prefix
– Lowest Administrative Distance
– Static default route
Administrative Distances: (Smallest Administrative Distance wins)

	Protocol	Distance
Connected	0	
Static (next hop)	1	
BGP External	20	
OSPF Internal	90	
EIGRP	110	
ISIS	115	
RIP v1/v2	120	
EIGRP External	170	
BGP Internal	200	